

WEEK 1 PROJECT REPORT

Wealth vs Health – Does Money Buy a Longer Life?

1. Introduction

During the first week of our project, our team discussed different real-world problems that could be explored using data science. After discussing various ideas, we selected a problem related to wealth, education, healthcare and life expectancy.

Our main question was whether a country's economic wealth, especially GDP, is the main reason for people living longer, or whether factors such as education and immunization have a greater influence on life expectancy.

The topic was selected because there are noticeable differences in life expectancy between countries, even when they belong to the same time period. The project abstract highlights the example of Japan, with a life expectancy of around 84 years, compared with Sierra Leone at around 51 years.

2. Problem Statement

The problem identified by our team is to study the relationship between economic wealth and life expectancy and find out which factors have a stronger connection with the health of a country's population.

Main question: Does having more money automatically lead to a longer life, or do education and healthcare-related factors play a more important role?

3. Dataset Selection

For the project, our team selected a country-level dataset containing information related to economic, social, environmental and health indicators. The project materials describe the main source as the WHO Global Health Observatory dataset, covering countries over the 2000–2015 period and containing indicators related to health, economy and society.

The dataset used for the project contains country-level information such as GDP, GDP per capita, life expectancy, infant mortality, population, population growth, school enrollment, employment, internet users, urban population, fertility, CO₂ emissions, imports and exports, and region.

4. Initial Understanding of the Problem

While discussing the project, we understood that GDP alone may not give a complete picture of the health of a population. A country can have a high income but still face health-related problems if education, vaccination and healthcare access are not strong.

The project abstract also points out that schooling and immunization showed stronger relationships with life expectancy than GDP in the planned analysis. This gave our team a clear direction for the project. Instead of looking only at GDP, we decided to consider different factors together.

5. Work Done During Week 1

- Discussed different possible problem statements.
- Selected the topic “Wealth vs Health – Does Money Buy a Longer Life?”
- Defined the main research question for the project.
- Selected a suitable country-level dataset.

- Examined the variables available in the dataset.
- Discussed the possible relationship between GDP, education, healthcare and life expectancy.
- Prepared a PowerPoint presentation explaining the project idea.
- Prepared the project abstract.
- Discussed the future direction of the project and the type of analysis that can be performed on the dataset.

6. PPT Preparation

As part of Week 1, our team created a PPT to present the project idea. The presentation covers the problem background, dataset, objectives, analytical workflow, important variables and the expected direction of the project.

The PPT also explains that the project will compare factors such as GDP, schooling and immunization to understand their relative relationship with life expectancy. The presentation helped us organize our idea and made it easier to explain the project to others.

7. Project Abstract

We also prepared an abstract for the project. The abstract summarizes the main purpose of the project, dataset, methodology and expected findings. It focuses on determining whether economic wealth is the main driver of life expectancy and comparing it with educational and health-related factors.

It also describes the planned use of techniques such as correlation analysis, PCA and feature selection to study the important factors affecting life expectancy.

8. Expected Outcome

At the end of Week 1, our main goal was not to complete the entire analysis, but to establish a clear foundation for the project.

- Whether GDP has a strong relationship with life expectancy.
- How education is related to life expectancy.
- Whether immunization and other health indicators have a significant impact.
- Differences between countries and regions.
- Which factors should receive more attention when improving population health.

The project materials indicate that the final project is intended to include data analysis and an interactive Power BI dashboard for exploring country-level health factors.

9. Conclusion

Week 1 mainly focused on choosing the problem, selecting the dataset and preparing the basic project materials. Our team finalized the topic “Wealth vs Health – Does Money Buy a Longer Life?” and selected a dataset containing economic, educational, health and social indicators.

We also completed the initial PPT and project abstract. This week helped us get a clear idea of what we want to investigate and how we can use data science techniques to study the problem.

In the coming weeks, we plan to work on data cleaning, exploratory data analysis, visualization and statistical analysis to get meaningful results from the selected dataset.